Practice workbook · Finding the length between two points

Using the distance formula to find a length between two points, often leaving the answer as a simplified surd, and using those lengths to compare sides or form a ratio.

10 question parts: 1 worked-example retry and 9 further practice parts.
Name: _____ Class: _____ Date: _____

Practise one question type at a time. Attempt the part inside the red boundary; the rest of the question and any figures are kept for context. Write your working in the answer space.

The first question is a retry of the website worked example, without its solution. Check your answers online with 'Check my answer', or ask your teacher for the answer booklet.

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Type ID: FPM-COO-01

1. Question 8 · part a · 4 marks

2019 JUN · Paper 02

Worked-example retry · Attempt part a only.

8 The point A has coordinates (2, 6), the point B has coordinates (6, 8) and the point C has coordinates (4, 2).

(a) Find the exact length of

(i) AB (ii) BC (iii) AC

(4)

(b) Find the size of each angle of triangle ABC in degrees.

(3)

The points A, B and C lie on a circle with centre P.

(c) Find the coordinates of P.

(2)

(d) Find the exact length of the radius of the circle in the form $\sqrt{a}$, where a is an integer.

(2)

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P 5 8 3 7 3 A 0 1 8 3 2

1. Question 8 · part a · 4 marks · Working and answer

Working and answer

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2. Question 5 · part b · 2 marks

2020 NOV · Paper 01

Further practice · Attempt part b only.

5 The points P, Q, R and S have coordinates (4, 7), (3, 0), (10, 1) and (11, 8) respectively.

(a) Show, by calculation, that the lines PR and QS are perpendicular.

(3)

(b) Find the exact lengths of (i) PR (ii) QS

(2)

(c) Find the area of the quadrilateral $PQRS$

(2)

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2. Question 5 · part b · 2 marks · Working and answer

Working and answer

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3. Question 2 · part a · 2 marks

2025 JUN · Paper 01

Further practice · Attempt part a only.

2 The point A has coordinates $(3, 2)$, the point B has coordinates $(8, 3)$ and the point C has coordinates $(4, 7)$

(a) Show that ABC is an isosceles triangle.

(2)

The midpoint of BC is M

(b) Find an equation of the line that passes through A and M

Give your answer in the form $y = mx + c$

(3)

The points A, C and D are collinear such that $AD = kAC \quad (k > 1)$

Given that $\angle ABD = 90^\circ$

(c) find the coordinates of D
Show your working clearly.

(8)

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3. Question 2 · part a · 2 marks · Working and answer

Working and answer

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4. Question 2 · part b · 2 marks

2022 JAN · Paper 02 (regional R)

Further practice · Attempt part b only.

2 The point A has coordinates $(-7, -1)$ and the point B has coordinates $(3, 4)$

(a) Find an equation of the line that passes through A and B
Give your answer in the form $ax + by + c = 0$ where a, b and c are integers.

(3)

The point C has coordinates $(-3, 7)$

Given that k is a constant such that $AB = kAC$

(b) find the value of k

(2)

The point D has coordinates $(3, p)$ where p is a constant.

Given that CD is perpendicular to AB

(c) find the value of p

(3)

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4. Question 2 · part b · 2 marks · Working and answer

Working and answer

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5. Question 8 · part d · 2 marks

2019 JUN · Paper 02

Further practice · Attempt part d only.

8 The point A has coordinates (2, 6), the point B has coordinates (6, 8) and the point C has coordinates (4, 2).

(a) Find the exact length of

(i) AB (ii) BC (iii) AC

(4)

(b) Find the size of each angle of triangle ABC in degrees.

(3)

The points A, B and C lie on a circle with centre P.

(c) Find the coordinates of P.

(2)

(d) Find the exact length of the radius of the circle in the form $\sqrt{a}$, where a is an integer.

(2)

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5. Question 8 · part d · 2 marks

2019 JUN · Paper 02

Further practice · Attempt part d only.

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Question 8 continued

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5. Question 8 · part d · 2 marks

2019 JUN · Paper 02

Further practice · Attempt part d only.

Question 8 continued

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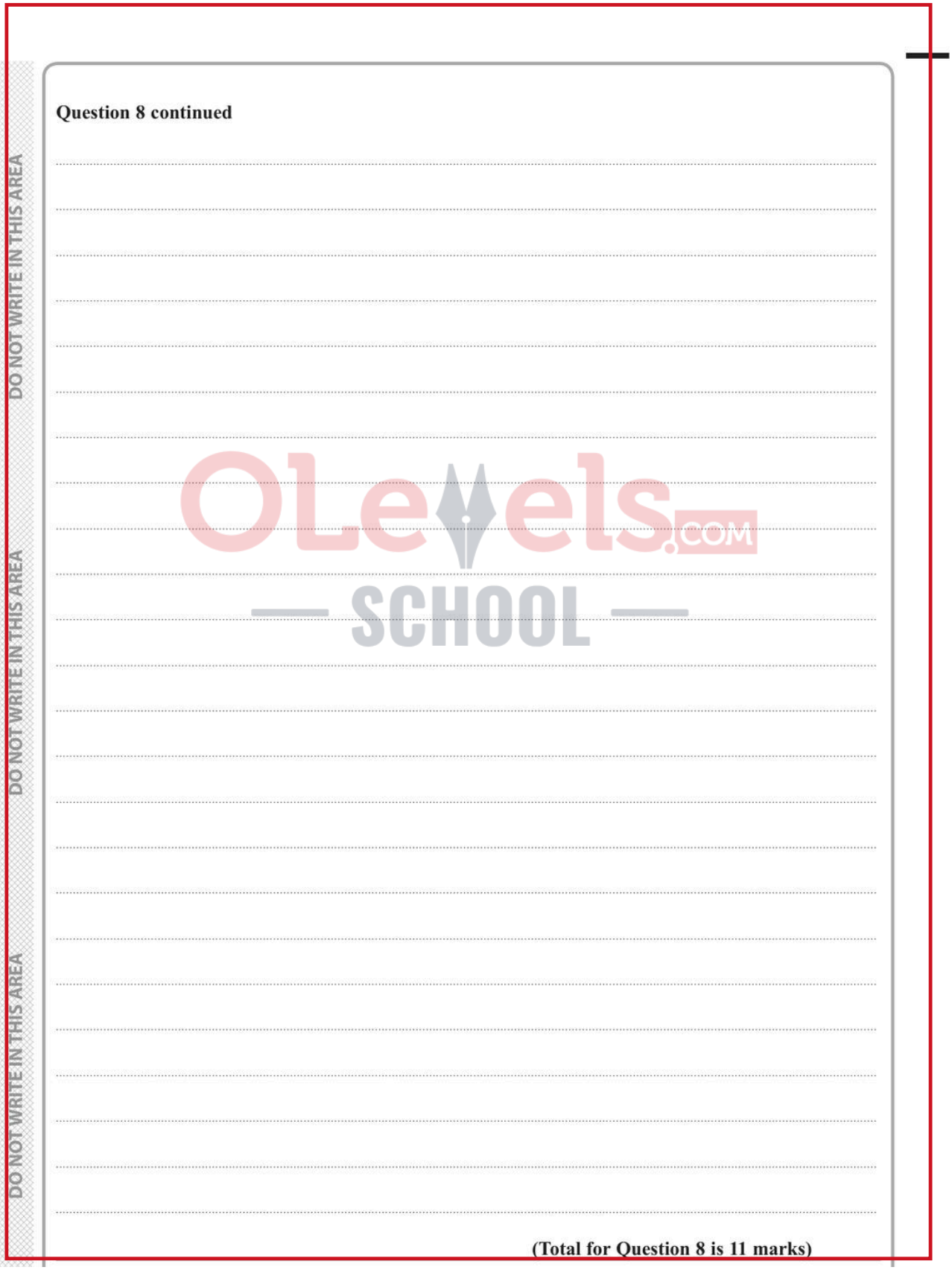
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5. Question 8 · part d · 2 marks

2019 JUN · Paper 02

Further practice · Attempt part d only.



Question 8 continued

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(Total for Question 8 is 11 marks)

5. Question 8 · part d · 2 marks · Working and answer

Working and answer

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6. Question 6 · part c · 3 marks

2024 JUN · Paper 01

Further practice · Attempt part c only.

6 The line l passes through the point A with coordinates $(-2, 2)$ and the point B with coordinates $(3, 12)$
The point C with coordinates (p, q) lies on l such that $AC : CB = 3 : 2$

(a) Find the value of p and the value of q

(2)

The line k is perpendicular to l and passes through the point C

(b) Show that an equation of k is $2y + x - 17 = 0$

(4)

The line k crosses the x-axis at the point D

(c) Find the exact length of CD

(3)

The point X with coordinates (m, n) lies on l such that

$$\text{area of triangle } DXC = 80 \text{ units}^2$$

Given that $m > 0$

(d) find the value of m and the value of n

(7)

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P 7 6 5 0 6 A 0 1 2 3 2

6. Question 6 · part c · 3 marks · Working and answer

Working and answer

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7. Question 4 · part d.i · 0 marks

2021 JUN · Paper 02

Further practice · Attempt part d.i only.

4 The point A has coordinates $(-4, -10)$ and the point B has coordinates $(3, 11)$
The line l passes through A and B.

(a) Find an equation of l.

(2)

The point P lies on l such that $AP : PB = 3 : 4$

(b) Find the coordinates of P.

(2)

The point Q with coordinates (m, n), where $m < 0$, lies on the line through P that is perpendicular to l.

Given that the length of PQ is $\sqrt{10}$

(c) find the coordinates of Q.

(6)

The point R has coordinates $(-11, -21)$

(d) Show that

(i) AB and RQ are equal in length,

(ii) AB and RQ are parallel.

(4)

(e) Find the area of the quadrilateral $ABQR$.

(2)

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7. Question 4 · part d.i · 0 marks · Working and answer

Working and answer

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8. Question 11 · part d.i · 0 marks

2021 NOV · Paper 02

Further practice · Attempt part d.i only.

11 The points A and B have coordinates $(-3, -5)$ and $(7, 5)$ respectively.

(a) Find an equation for the line AB

(2)

The point C has coordinates $(p, 1)$ where $p < 0$

Given that AC and BC are perpendicular,

(b) prove that $p = -5$

(7)

The point D, where BCD is a straight line, is such that C divides BD in the ratio $4:3$

(c) Find the coordinates of D

(2)

(d) (i) Find the exact length of AC

(ii) Hence, or otherwise, find the area of triangle ABD

(4)

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8. Question 11 · part d.i · 0 marks · Working and answer

Working and answer

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9. Question 8 · part b · 2 marks

2023 JUN · Paper 02

Further practice · Attempt part b only.

8 The points A and B have coordinates $(-6, 8)$ and $(12, 2)$ respectively.

(a) Find an equation of the straight line passing through A and B in the form $ax + by + c = 0$, where a, b and c are integers to be found.

(3)

(b) Find the exact length of AB

(2)

The point X with coordinates (m, n) lies on AB such that $AX:XB = 1:2$

(c) Find the value of m and the value of n

(2)

The line L passes through the point X and is perpendicular to AB
The point C with coordinates (p, q) lies on L where $p > 0$ and $q > 0$

Given that AB is a diameter of a circle and C also lies on the circumference of the circle,

(d) find

(i) the exact value of p

(ii) the exact value of q

(7)

(e) Find the exact area of triangle ABC

(3)

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9. Question 8 · part b · 2 marks · Working and answer

Working and answer

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10. Question 2 · part d · 4 marks

2023 JAN · Paper 01

Further practice · Attempt part d only.

2 The point A has coordinates $(-5, 3)$, the point B has coordinates $(4, 0)$ and the point C has coordinates $(-1, 5)$.

The line l passes through C and is perpendicular to AB.

(a) Find an equation of l.
Give your answer in the form $ax + by + c = 0$ where a, b and c are integers.

(4)

The line l intersects AB at the point D.

(b) Show that the coordinates of D are $(-2, 2)$.

(3)

(c) Show that l is not the perpendicular bisector of AB.

(2)

(d) Find the value of $\tan \angle ABC$.
Give your answer in its simplest form.

(4)

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10. Question 2 · part d · 4 marks

2023 JAN · Paper 01

Further practice · Attempt part d only.

Question 2 continued

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10. Question 2 · part d · 4 marks

2023 JAN · Paper 01

Further practice · Attempt part d only.

Question 2 continued

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10. Question 2 · part d · 4 marks

2023 JAN · Paper 01

Further practice · Attempt part d only.

Question 2 continued

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(Total for Question 2 is 13 marks)

10. Question 2 · part d · 4 marks · Working and answer

Working and answer

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